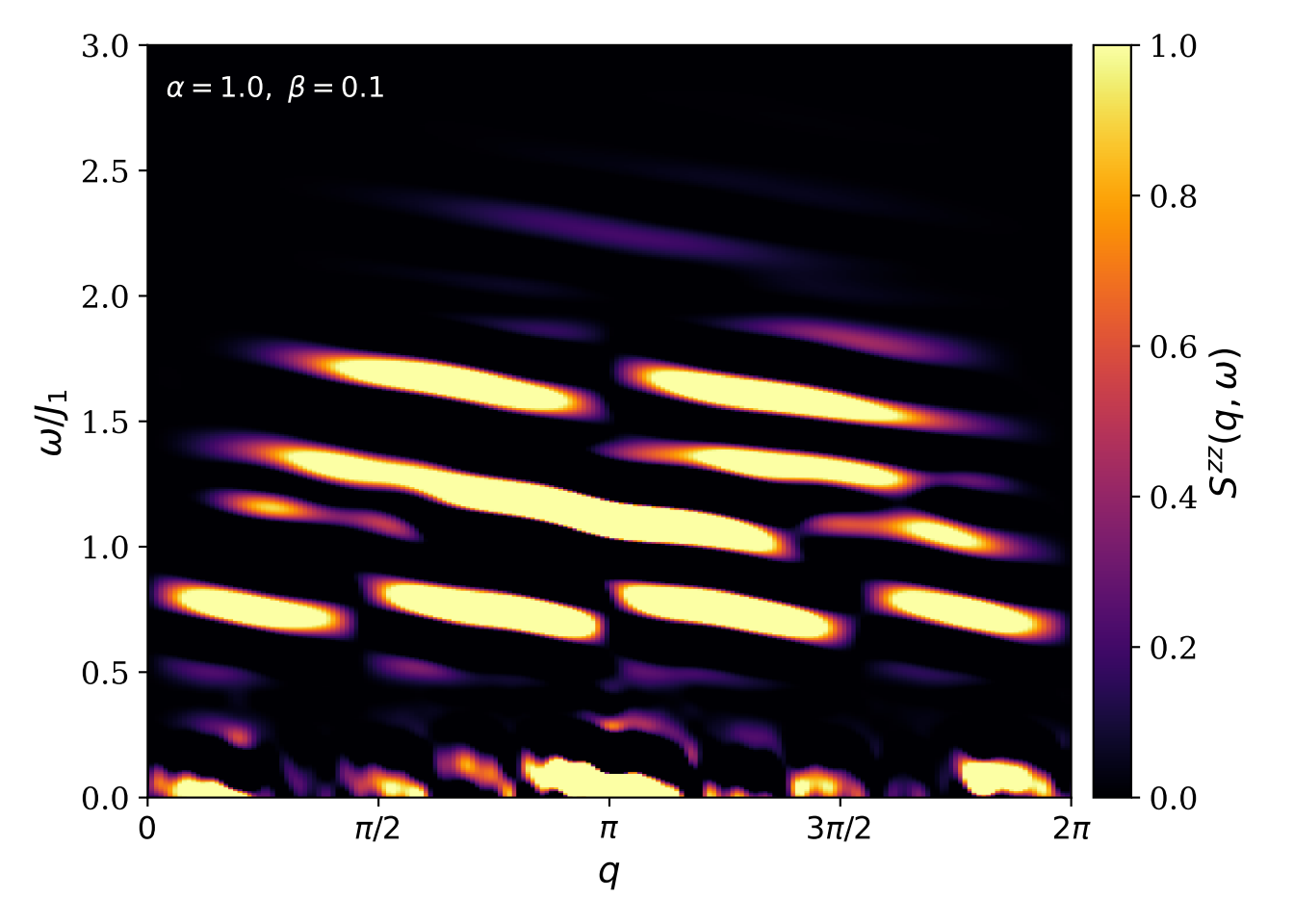

$\alpha = 1.0,\ \beta = 0.1$
3.0
2.5
2.0
1.5
1.0
0.5
0.0
ω/J_1
0
$\pi/2$
π
$3\pi/2$
2π
q
1.0
0.8
0.6
0.4
0.2
0.0
$S^{zz}(q,\omega)$